
Lens-LeJEPa: Learning the Right Invariances for Strong Gravitational Lenses

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Abstract

1 Strong gravitational lenses probe small-scale dark matter structure, but labelled
2 simulations are expensive and labelled real lenses barely exist. Self-supervised
3 pretraining should help, yet its usual augmentations are physically wrong here:
4 zooming a lens image rescales the Einstein radius. **Lens-LeJEPa** replaces them
5 with two lensing priors on a teacher-free joint-embedding objective, exact D_4
6 correspondence between image patches and consistency of arc-weighted descriptors
7 pooled in lens-centred rings. One encoder, pretrained on 87.5k unlabelled images
8 and then frozen, serves three tasks through low-rank adapters. Under a matched
9 budget the priors add 11.6 accuracy points at 100 labels per class and beat the
10 previous lensing-specific JEPa by 13 to 30 points, while rank-stabilized LoRA
11 matches full fine-tuning with 11% of the weights. We also report where the
12 approach loses, on axion-mass regression, and argue both outcomes follow from
13 the same invariance.

14 1 Introduction

15 A massive galaxy between us and a distant source bends its light into arcs whose shape depends
16 on how the lens mass is distributed, making strong lenses one of the few probes of dark matter
17 below galactic scales [Vegetti et al., 2010]. Dark matter models leave different marks: cold dark
18 matter adds discrete subhaloes, while an ultra-light axion-like particle forms vortex-like defects
19 [Hu et al., 2000, Alexander et al., 2020a]. Supervised networks separate these well on simulations,
20 but need large label sets a simulation can only produce once it has assumed a dark matter model,
21 and Euclid [Laureijs et al., 2011] will deliver many lens images and no labels. That is the setting
22 self-supervised pretraining exists for, yet generic recipes transfer poorly here: their strength comes
23 from augmentations that are physically wrong, since zooming rescales the apparent Einstein radius
24 θ_E , and θ_E fixes the mass enclosed within it through $M \propto \theta_E^2$. A zoom-invariant encoder is therefore
25 trained to discard a mass measurement.

26 Our contributions:

- 27 • **Lens-LeJEPa**, a teacher-free objective with two lensing priors, exact D_4 patch-token correspon-
28 dence and arc-weighted ring consistency, computed from the image alone with no lens model and
29 no labels.
- 30 • **A controlled comparison** in which those priors are the only difference between two otherwise
31 identical runs; they help at every label budget and beat Lens-JEPa by 13 to 30 accuracy points.
- 32 • **An adapter study** over seven families and a rank sweep, and **one frozen encoder on three tasks**,
33 reporting where it wins, where it loses, and one mechanism explaining both.

34 2 Physics and data

35 The public DeepLense simulations [Toomey, 2021], generated with lenstronomy [Birrer and
36 Amara, 2018, Birrer et al., 2021], place a sheared isothermal elliptical lens in front of a Sérsic
37 source (Appendix B). The three classes differ only in the substructure added to the main halo, **axion**

(vortex-like defects), **CDM** (point-mass subhaloes) and **no substructure**, and that difference appears as fine perturbations of the arcs rather than a change in the overall ring. **Model I** (150^2 px, 87.5k images) is our *only* pretraining source and none of its labels are used; **Model II** (Euclid-like) and **Model III** (HST-like), both 64^2 px with 15k test images, are downstream only, so every result is cross-instrument. Axion images carry the simulated particle mass, 10^{-24} to 10^{-22} eV, the regression target $\log_{10}(m/\text{eV})$. Two properties drive our design. The lens is centred and the simulated point spread function is isotropic, so a 90° rotation or reflection is an equally valid observation of the same system, with class, θ_E and mass unchanged. And the signal sits in thin arcs near θ_E .

3 Method

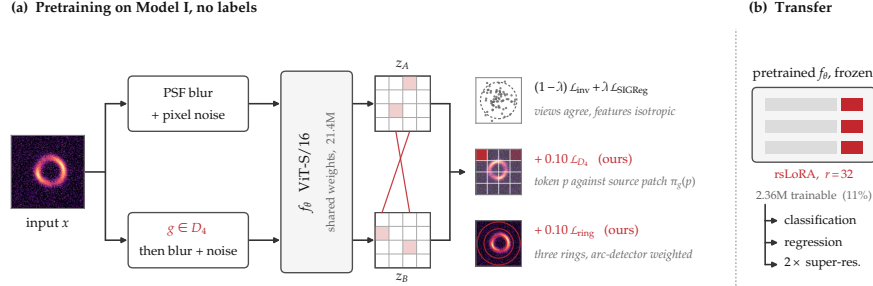


Figure 1: **Lens-LeJEPA**. (1) One ViT-S/16 encodes two views of an unlabelled Model I image differing only by lens-safe perturbations and a D_4 transform; two of the three summed terms are ours. (2) The quantities those priors use, computed with the training code’s own formulas on a real image, none of which needs a label or a lens model. (3) The encoder is frozen and adapted with rsLoRA per task.

Base objective. We build on LeJEPA [Balestriero and LeCun, 2025], which learns from two views with no teacher, no moving average and no stop-gradient. Both pass through the *same* ViT-S/16 [Dosovitskiy et al., 2021], giving patch-token maps $z_A, z_B \in \mathbb{R}^{P \times d}$ over the $P = 10 \times 10$ grid, which mean-pooling and a small projector reduce to vectors u_A, u_B . One term pulls the projections together; a sketched isotropy term (SIGReg) keeps their distribution near an isotropic Gaussian, which is what prevents collapse without a teacher, in the spirit of variance-covariance regularisation [Bardes et al., 2022]:

$$\mathcal{L}_{\text{base}} = (1 - \lambda) \underbrace{\frac{1}{V} \sum_v \|u_v - \bar{u}\|^2}_{\mathcal{L}_{\text{inv}}} + \lambda \mathcal{L}_{\text{SIGReg}}, \quad \lambda = 0.02, \quad (1)$$

with $\bar{u}$ the view mean and gradients through both branches. Crucially $\mathcal{L}_{\text{base}}$ sees only the pooled vectors; our priors act on the token maps, where the image-plane geometry survives.

Lens-safe views. A view must be an image the telescope could plausibly have returned for the *same* system. That admits a Gaussian blur of at most one native pixel, for point-spread-function variation, and low-amplitude detector noise, and excludes what contrastive learning normally relies on: crop, zoom and rescale all change the apparent θ_E while the system is fixed, so an encoder trained to ignore them is trained to ignore a mass measurement [Chang et al., 2023]. Translation is excluded separately, since it moves the lens off centre and both priors are defined relative to that centre.

Prior 1: exact D_4 correspondence. The one transformation that *is* safe is a change of detector orientation. We draw g from the dihedral group D_4 and apply it before forming the second view. Because g acts on a centred grid whose side is an exact multiple of the patch size, it maps whole patches onto whole patches, inducing an exact permutation π_g of patch indices, proved and verified numerically in Appendix E. That lets us compare tokens patch by patch rather than imposing one global invariance:

$$\mathcal{L}_{D_4} = \text{smooth}_{L_1}(\text{LN}(z_B), \text{LN}(z_A[\pi_g])), \quad (2)$$

where LN is layer normalisation and $z_A[\pi_g]$ reindexes the canonical map. The term vanishes exactly when the encoder is D_4 -equivariant, so it asks the network to learn the equivariance group-equivariant architectures hard-code [Cohen and Welling, 2016, Bronstein et al., 2021], without changing the architecture.

Prior 2: arc-weighted ring consistency. Patch correspondence constrains *where* information sits, not *what* the arcs look like, and the substructure signal is a faint perturbation of arc morphology. We therefore pool each view’s tokens into $K = 3$ lens-centred rings, weighting every patch by a label-free arc detector $w(x)$ from the image gradient plus Laplacian magnitude, normalised to unit mean, so patches on a bright curved arc dominate. With R_k the ring- k patch set,

$$\phi_k(z, x) = \frac{\sum_{p \in R_k} w_p(x) z_p}{\sum_{p \in R_k} w_p(x)}, \quad \mathcal{L}_{\text{ring}} = \text{smooth}_{L_1}(\text{LN}(\phi(z_B, x_B)), \text{LN}(\phi(z_A, x_A))). \quad (3)$$

Since every element of D_4 preserves distance from the centre, π_g maps each ring to itself (Appendix E), so ϕ is invariant by construction and matching it constrains arc morphology at each radius instead of restating the orientation constraint. The priors are therefore complementary, and the objective is $\mathcal{L} = \mathcal{L}_{\text{base}} + 0.10 \mathcal{L}_{D_4} + 0.10 \mathcal{L}_{\text{ring}}$.

Transfer and protocol. The encoder is then frozen and every linear projection receives a rank-stabilized LoRA adapter [Kalajdziewski, 2023], which scales the update by $\alpha/\sqrt{r}$ rather than α/r so its magnitude does not shrink as rank grows; at $r=32$ that is 2.36M parameters, 11% of the encoder, and one backbone serves all three tasks. Both encoders are 12-block ViT-S/16 at 160^2 , 100 epochs on Model I, identical seed and checkpoint policy; downstream runs use 50 epochs and the best-validation checkpoint, and every number is held-out (Appendix C). I-JEPA and the Lens-JEPA family are $3\times$ -budget references at 300 epochs.

4 Results

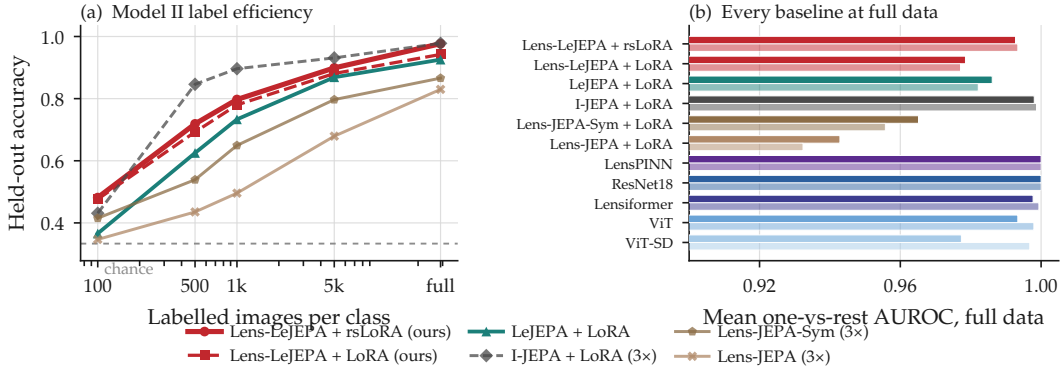


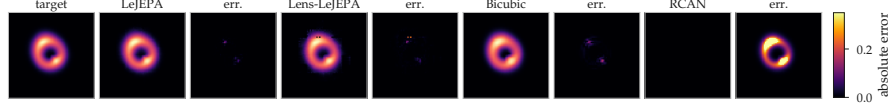
Figure 2: **Substructure classification.** (a) Label efficiency on Model II; every curve shares the frozen-encoder, rank-32 adapter protocol, so vertical gaps measure the pretraining objective alone. (b) Every baseline at full data, mean one-vs-rest AUROC, with the darker bar Model II and the lighter bar Model III. The complete matrix at every budget is Appendix Table 6, and per-class values are Appendix Table 7.

4.1 Substructure classification

Against the identical objective without the priors (Figure 2a), they are worth +11.6 accuracy points at 100 labels per class, +9.4 at 500 and +5.1 at full data, the AUROC gain being largest where labels are scarcest (+13.5 at 100). Against Lens-JEPA the margin is 13 to 30 points despite its $3\times$ budget, and a soft *global* D_4 loss (Lens-JEPA-Sym) recovers only part of it. At full data (Figure 2b) Lens-LeJEPA reaches AUROC 0.9927 on Model II and 0.9934 on Model III, ahead of the priors-free objective (0.9861, 0.9822) and the whole Lens-JEPA family, and passes ViT-SD on both metrics on Model II. On AUROC every other supervised model leads slightly. The case for a foundation model is label efficiency and multi-task reuse, not full-data accuracy.

4.2 Axion mass regression

Estimating a continuous physical quantity is harder. The frozen encoder is usable, 0.100 dex MAE and $R^2 = 0.937$ on Model II, the best self-supervised result on both datasets (Figure 3, lower left), but a supervised ResNet18 reaches 0.021 dex, about $4.8\times$ better. This follows from the objective, not a tuning failure. A heavier axion has a shorter de Broglie wavelength, so its vortex pattern is finer and lower in contrast, while our objective explicitly asks for *invariance* to blur and noise acting on that same spatial band. The error profile agrees: it roughly quadruples from the lowest to the highest mass decile (Appendix Figure 9), that is, towards finer patterns.



Axion mass (MAE, dex $\downarrow$)			2 $\times$ SR (PSNR, dB $\uparrow$)		
Method	M-II	M-III	Method	M-II	M-III
Lens-LeJEPA + rsLoRA (ours)	<u>0.100</u>	<u>0.090</u>	LeJEPA + LoRA	48.32	45.38
Lens-LeJEPA + LoRA (ours)	<u>0.149</u>	<u>0.109</u>	Lens-LeJEPA + LoRA (ours)	<u>44.70</u>	<u>42.50</u>
LeJEPA + LoRA	0.121	0.122	Bicubic	41.07	40.27
LensPINN (supervised)	0.042	0.035	RCAN	19.49	19.01
ResNet18 (supervised)	0.021	0.019	EDSR	<u>42.60</u>	<u>41.90</u>
			SRGAN	<u>42.10</u>	<u>41.40</u>
			SRResNet	<u>41.80</u>	<u>41.20</u>

Figure 3: **Downstream tasks.** Top: 2 $\times$ super-resolution on held-out Model II; reconstruction/error maps use a shared scale. SR uses bicubic 64 $\rightarrow$ 32 inputs and measures reconstruction, not physical resolution recovery. Best bold, ours underlined.

4.3 Synthetic 2 $\times$ super-resolution

Here the frozen encoders do well (Figure 3, lower right). A light token-to-patch decoder reaches 48.32 dB on Model II and Lens-LeJEPA 44.70 dB, 7.24 and 3.62 dB above bicubic, with residual error on the brightest arc pixels. RCAN [Zhang et al., 2018], trained from scratch at the identical budget, fails to converge, so it is budget-matched rather than the best achievable RCAN. The task is synthetic, and here the priors *cost* 3.6 dB: the mirror image of the regression result.

4.4 Ablations

Table 1: **Adaptation capacity.** Held-out accuracy and per-step cost (batch 64, RTX A4500). Best values bold, ours underlined. Lens-LeJEPA has no frozen-probe run, so the probe comparison uses the reference encoder.

	Train.	ms /	Model II accuracy by label budget					Model III
Adaptation	(M)	step	100	500	1k	5k	full	full
<i>Lens-LeJEPA encoder (ours)</i>								
LoRA $r=32$	2.36	121	0.4763	0.6927	0.7797	0.8809	0.9423	0.9453
rsLoRA $r=32$ (ours)	2.36	122	0.4815	0.7188	0.7972	0.8989	0.9768	0.9744
Full fine-tuning	21.39	102	0.4851	0.7150	0.8084	0.8979	0.9751	0.9776
<i>Reference I-JEPA encoder: what an adapter buys over a probe</i>								
Frozen linear probe	0.001	35	0.3848	0.4121	0.4351	0.4884	0.6002	0.6235
LoRA $r=32$	2.36	121	0.4311	0.8461	0.8963	0.9311	0.9780	0.9905

The two priors. Removing both gives exactly the LeJEPA curve in Figure 2: that costs 5.1 to 11.6 accuracy points on Model II and 5.3 on Model III, and lowers the pooled effective-rank fraction from 0.213 to 0.164 (Appendix Figure 5). Replacing exact patch correspondence with a global symmetry loss recovers only part of the gap, because π_g is exact and a pooled loss discards that correspondence (Appendix E).

Adaptation capacity. A frozen probe is far too weak: on the reference encoder it reaches 0.6002 at full data where LoRA reaches 0.9780 (Table 1). Full fine-tuning buys almost nothing over rsLoRA, at most 1.2 points and losing on three of six settings, while training 9 $\times$ more parameters. rsLoRA’s saving is in trainable parameters and optimiser state, *not* step time: at batch 64 it runs at 122 ms/step against 102 ms. Across seven adapter families (Appendix Table 4) rsLoRA is best on all six.

5 Discussion and conclusion

One mechanism explains all three results: the lensing priors enforce invariance to detector orientation and absolute patch position, which helps semantic tasks but hurts detail-sensitive ones. Classification benefits because substructure class should be rotation- and position-invariant, gaining 5–12 points over the same objective without priors. Super-resolution instead loses 3.6 dB because reconstruction requires the pixel-level detail these invariances discard, while mass regression remains 4.8 $\times$ worse than supervised as blur invariance suppresses fine texture. The pooled effective-rank drop from 0.213 to 0.164 reflects the same loss of information in the representation, which a rank-32 adapter cannot recover. Thus, the geometry-aware encoder is preferable for classification and lens discovery, while the encoder without these priors is better suited to continuous physical quantities. Appendix H discusses limitations, chiefly the reliance on simulation.

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A Related work

Substructure from lens images. Machine learning here began with supervised classifiers [Alexander et al., 2020a] and unsupervised clustering [Alexander et al., 2020b]. The strongest current baselines fold the lens equation into the network: Lensformer [Veloso et al., 2023] adds a branch approximating the lensing potential and LensPINN [Ojha et al., 2024] extends it with physics-informed preprocessing. Both are supervised, and both lead us at full data.

Self-supervised lensing models. Contrastive pretraining has been used to mine surveys for lens candidates [Stein et al., 2022], and LenSiam [Chang et al., 2023] made the observation this paper builds on: crop and zoom change the Einstein radius, destroying the quantity a lensing model should measure. **Lens-JEPA** [Rishi et al., 2025] is the closest work and our main comparison. It extends I-JEPA [Assran et al., 2023], which predicts masked target blocks in latent space with a moving-average teacher, by adding a physics encoder guided by the lens equation. A concurrent masked-autoencoder study pairs classification with super-resolution here [Prasha et al., 2025].

What we do differently. All of the above put the physics, or the prediction machinery, into the *architecture*. We put it into the *objective*, on the teacher-free LeJEPA formulation [Balestrierio and LeCun, 2025], so each prior is one additive term whose removal gives an exact ablation, and the encoder stays an ordinary ViT with no teacher, moving average or mask sampler.

B Data and physics details

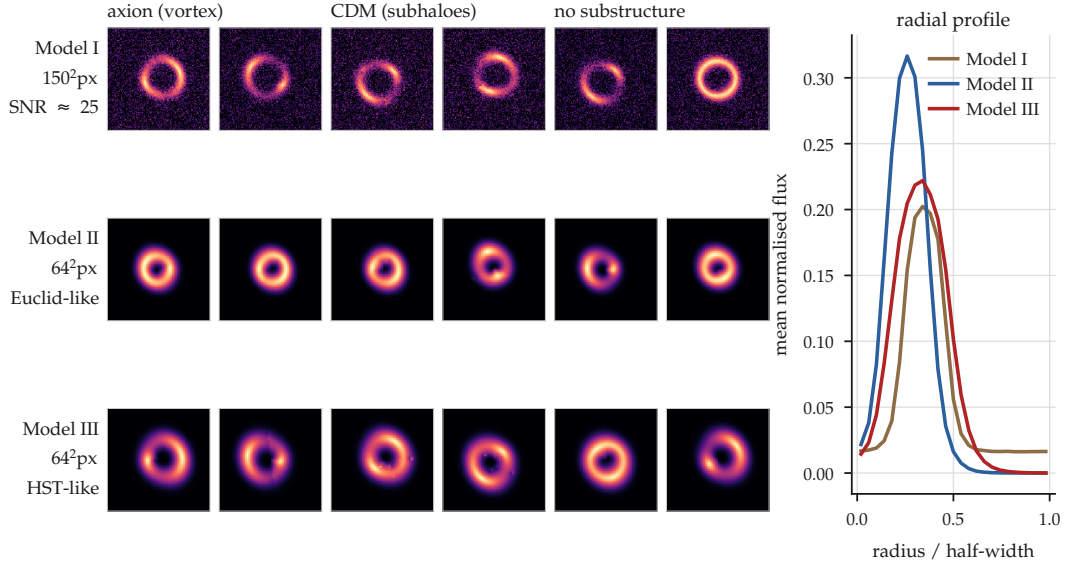


Figure 4: **The three DeepLense datasets.** Two held-out examples per class per dataset, shown with an arcsinh stretch. The class difference is a perturbation of the arc, not a change of the overall ring: axion images carry vortex-like interference structure, CDM images carry discrete subhalo perturbations, and the no-substructure images have a smooth arc. The right panel gives the azimuthally averaged radial profile of each dataset. Model I is noisier and broader, Model II (Euclid-like) is the most concentrated, and Model III (HST-like) sits between them; this is the domain shift the frozen encoder must survive, since pretraining only ever sees Model I.

Simulation setup and attribution. All images come from the public DeepLense simulations, produced by the DeepLenseSim pipeline [Toomey, 2021] on top of lenstronomy [Birrer and Amara, 2018, Birrer et al., 2021]; the substructure models they implement are those of Alexander et al. [2020a,b]. We cite the pipeline and the software separately because the datasets have no paper of their own, and we follow lenstronomy’s own attribution policy in citing both software references. The lens is a sheared isothermal elliptical mass distribution and the background source follows a Sérsic profile. Substructure is added to the base halo to create the three classes. Model I applies a Gaussian point spread function with Gaussian and Poisson noise at signal-to-noise ≈ 25 at 150×150 pixels; Model II applies Euclid-like observing characteristics at 64×64 ; Model III applies HST-like characteristics at 64×64 .

Class counts. Model I: 28,897 axion / 29,772 CDM / 28,856 no-substructure (87,525 unlabelled pretraining images). Model II: 29,896 / 29,759 / 29,449 training and 5,000 per class held-out test. Model III: 29,812 / 29,584 / 30,000 training and 5,000 per class held-out test. Downstream validation is a stratified split of the training partition; the official test partitions are used once, for the final evaluation.

Why the priors are physically reasonable. The orientation of a lens on the sky is a matter of convention, the simulated point spread function is isotropic, and the lens sits at the centre of the frame. Applying an element of D_4 to the image is therefore equivalent to rotating or reflecting the detector, and produces an equally valid observation of the same system: the class, the Einstein radius θ_E and the axion mass are all unchanged. That is what makes exact patch correspondence a safe constraint rather than an assumption.

An image-plane crop, zoom or rescale is different in kind. It changes the *apparent* θ_E in pixels while the physical system is fixed, and θ_E determines the mass enclosed within it,

$$M(< \theta_E) = \frac{c^2}{4G} \frac{D_d D_s}{D_{ds}} \theta_E^2, \quad (4)$$

with D_d , D_s and D_{ds} the observer-lens, observer-source and lens-source angular diameter distances. Since $M \propto \theta_E^2$, a zoom-invariant encoder would be trained to discard a direct mass measurement, which is the opposite of what we want. Translation is excluded for a separate reason: it moves the lens off centre, and both the D_4 patch correspondence and the ring partition assume a centred lens.

Finally, the substructure signal is concentrated in thin arcs near θ_E , which is why a radius-organised, gradient-weighted descriptor is a useful summary, and why we weight patches by an image-gradient and Laplacian response rather than by a learned or label-dependent saliency.

Regression target. Only axion images carry a simulated mass, and it is the mass of the ultra-light *particle*, not of a halo: the stored values span 10^{-24} to 10^{-22} eV, the fuzzy dark matter scale [Hu et al., 2000]. The target is therefore $\log_{10}(m/\text{eV})$, the normaliser is fit on the training split alone, and splits are deterministic mass quantiles so the full target range appears in train, validation and test.

Super-resolution task definition. The native 64×64 image is the target. The input is that image after antialiased bicubic downsampling to 32×32 , resized back to 160×160 for the encoder. The task is therefore explicitly synthetic image reconstruction. The DeepLense release does not contain a paired low-/high-resolution physical simulation for Models II and III, so no physical de-lensing or resolution-recovery claim is made anywhere in this paper.

C Implementation and training details

Related DeepLense work. Beyond the supervised classifiers used as baselines here, unsupervised clustering of substructure has also been explored on this simulation family [Alexander et al., 2020b].

Concurrent work. A masked-autoencoder study on this benchmark also pairs dark matter classification with super-resolution [Prasha et al., 2025]. We do not tabulate it against our numbers: it uses a $16 \rightarrow 64$ super-resolution factor rather than our $32 \rightarrow 64$, and a different split, so the metrics are not comparable.

Encoders. Both self-supervised encoders are standard 12-block ViT-S/16 (384 dimensions, 6 heads, 21.43M parameters) at 160×160 input with 16×16 patches, giving a 10×10 token grid. Native ViT initialisation including residual-layer rescaling is preserved. The LeJEPa projector is a two-layer BatchNorm MLP to 256 dimensions; both views are projected in one batch so they share normalisation statistics.

Pretraining. 100 epochs on Model I only, effective batch 128, AdamW, peak learning rate 5×10^{-4} with 10 warm-up epochs and cosine decay, weight decay 0.05, bfloat16, 256 SIGReg slices, $\lambda = 0.02$, lens prior weights 0.10 each, three radial rings, arc-detector strength 0.75, blur probability 0.5 with $\sigma \leq 1$ native pixel, and noise standard deviation up to 0.03. Every epoch records feature standard deviation, effective-rank fraction, inter-view cosine, gradient norm and finiteness; a run that shows

three consecutive epochs below the collapse thresholds is rejected rather than used downstream. Neither reported run triggered this.

Downstream. Frozen encoder, adapters on all 48 linear projections, 50 epochs, batch 128, AdamW at 3×10^{-4} with weight decay 0.01, validation every 10 epochs, seed 42, and evaluation of the best-validation checkpoint only. Classification adds label smoothing 0.1; regression uses the normalised scalar target; super-resolution uses Charbonnier plus SSIM loss [Lai et al., 2019, Wang et al., 2004] with a token-to-patch decoder.

Adapter families. *LoRA* [Hu et al., 2022] scales by α/r ; *rsLoRA* [Kalajdziewski, 2023] by $\alpha/\sqrt{r}$; *DoRA* [Liu et al., 2024] decomposes the adapted weight into a learned row magnitude and a normalised direction; *LoRA+* [Hayou et al., 2024] gives the B matrices twice the learning rate of A ; *gated LoRA* follows the adaptive-budget idea of AdaLoRA [Zhang et al., 2023] with a learned sigmoid gate per rank component; *LoHA* uses the low-rank Hadamard product of Hyeon-Woo et al. [2022] as popularised by Yeh et al. [2024]; *layerwise LoRA* varies rank with block depth.

Baselines. The supervised models are a ResNet18 [He et al., 2016], a ViT [Dosovitskiy et al., 2021], ViT-SD, Lensiformer [Velôso et al., 2023] and LensPINN [Ojha et al., 2024], each trained end to end on the labelled downstream set. The two reference self-supervised encoders, I-JEPA [Assran et al., 2023] and the Lens-JEPA family [Rishi et al., 2025], were pretrained for 300 epochs under an earlier protocol: I-JEPA on a 21.5M three-block wide ViT, Lens-JEPA and Lens-JEPA-Sym on a 2.45M single-block lensing backbone. They are labelled as $3\times$ budget references throughout and are not matched comparisons, which is why the controlled claim in this paper is always against LeJEPA at our own budget.

Compute. Pretraining each encoder takes roughly 6 GPU-hours on a single RTX A4500; the complete set of downstream runs reported here is about 60 GPU-hours.

D Pretraining diagnostics

Both runs are single-seed, so the case that they are comparable rests on their training curves rather than on repetition. Figure 5 shows the nine quantities recorded every epoch. Three points matter. Neither run collapses: feature standard deviation rises and then plateaus, and the effective-rank fraction stays well above the 0.04 rejection threshold throughout, so neither checkpoint was accepted under a rescue. The two lensing terms converge rather than fight the base objective, falling by roughly two orders of magnitude, 1.9×10^{-2} to 1.5×10^{-4} for $\mathcal{L}_{D_4}$ and 6.7×10^{-3} to 1.3×10^{-4} for $\mathcal{L}_{\text{ring}}$, while the invariance and SIGReg terms follow the same trajectory in both runs. And the cost of the priors is visible directly in the representation: Lens-LeJEPA ends with a lower pooled effective-rank fraction, 0.164 against 0.213, and a lower feature standard deviation, 0.236 against 0.779. That is the invariance trade-off of Section 5, measured on the features rather than inferred from the downstream metrics.

E Why the patch permutation is exact

The D_4 term is worth stating precisely, because its value rests on a claim that is either exactly true or not true at all: that a D_4 transform of the image induces an *exact* permutation of patch indices, so the loss compares tokens of genuinely corresponding patches rather than merely encouraging a global invariance.

Setup. Let the image be $x \in \mathbb{R}^{N \times N}$ with $N = PG$, so that patch extraction $E : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}^{G^2 \times P \times P}$ cuts x into a $G \times G$ grid of $P \times P$ blocks, $E(x)_p$ being block $p = (i, j)$ in row-major order. Write $D_4 = \{r^k, r^k s : k = 0, \dots, 3\}$ for the dihedral group of the square, acting on the image about its centre: r is a quarter turn and s a reflection. In our setting $N = 160$, $P = 16$, $G = 10$.

Proposition 1 (Exact block permutation). *For every $g \in D_4$ there is a permutation π_g of $\{0, \dots, G^2 - 1\}$ such that, for all x and all p ,*

$$E(g \cdot x)_p = g \cdot E(x)_{\pi_g(p)}, \quad (5)$$

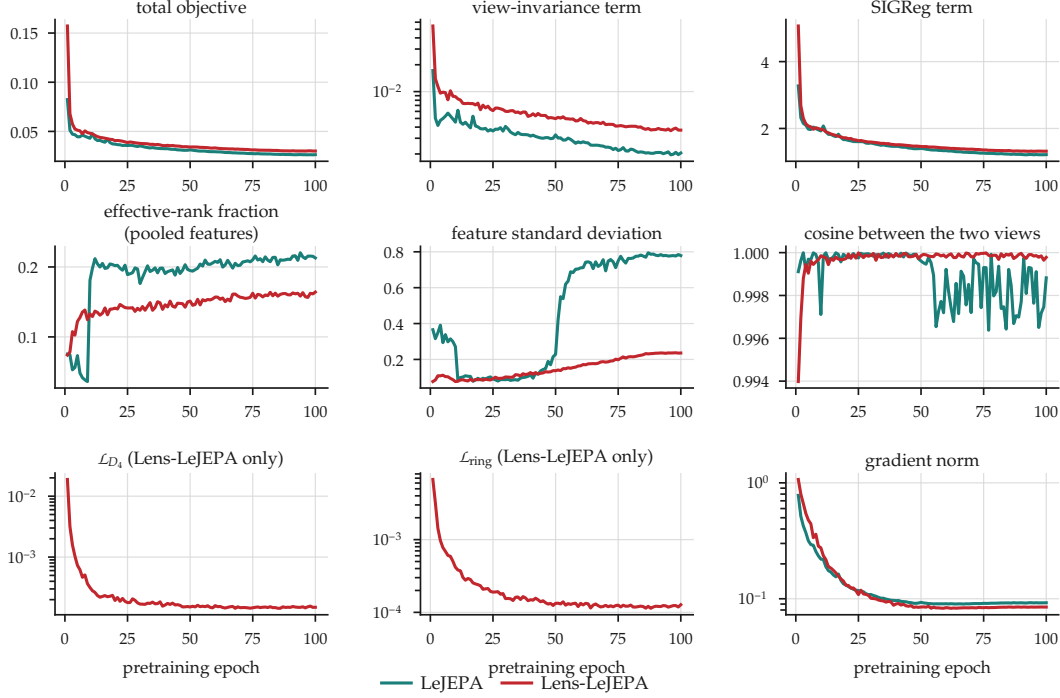


Figure 5: **Pretraining diagnostics over 100 Model I epochs.** Both runs are stable and neither collapses. The two lensing terms fall by roughly two orders of magnitude, so the constraints are satisfied rather than fought. The cost is visible in the middle row: Lens-LeJEPA ends with a lower pooled effective-rank fraction (0.164 vs 0.213) and lower feature standard deviation (0.236 vs 0.779) than the same objective without the priors. This is the quantitative version of the invariance trade-off discussed in the main text.

where the g on the right acts inside a single $P \times P$ block. Moreover $\pi : g \mapsto \pi_g$ is an injective group homomorphism $D_4 \rightarrow S_{G^2}$.

Proof. Index a pixel by $u = (u_1, u_2) \in \{0, \dots, N-1\}^2$ and decompose it uniquely as $u = Pq + r$ with block index $q \in \{0, \dots, G-1\}^2$ and offset $r \in \{0, \dots, P-1\}^2$. It suffices to check the two generators. For the quarter turn, $(r \cdot x)(u_1, u_2) = x(u_2, N-1-u_1)$. Substituting $u = Pq + r$ and using $N-1 = PG-1 = P(G-1) + (P-1)$ gives

$$N-1-u_1 = P((G-1)-q_1) + ((P-1)-r_1),$$

in which the first term is a multiple of P and the second lies in $\{0, \dots, P-1\}$. So this is again a valid block/offset decomposition, with block index $(q_2, (G-1)-q_1)$ and offset $(r_2, (P-1)-r_1)$. The block index depends only on q and the offset only on r : the action factors over the decomposition, which is exactly (5) with $\pi_r(q_1, q_2) = (q_2, (G-1)-q_1)$. The same substitution for the reflection $(s \cdot x)(u_1, u_2) = x(u_1, N-1-u_2)$ gives $\pi_s(q_1, q_2) = (q_1, (G-1)-q_2)$. Because both generators act on the block index alone, so does every word in them, and π is a homomorphism; it is injective because $\pi_g = \text{id}$ forces g to fix every block, hence to be the identity whenever $G \geq 2$. $\square$

Two consequences matter for the method. First, $\mathcal{L}_{D_4}$ is well posed: by (5), token p of the transformed view and token $\pi_g(p)$ of the canonical view see the same pixels up to the in-block action, so the loss is comparing physically corresponding patches, not an arbitrary spatial pairing. A global invariance loss on pooled features, as in Lens-JEPA-Sym, discards this correspondence, which is what the ablation in Section 4.4 measures. Second, the loss vanishes exactly on the set of D_4 -equivariant encoders, that is those with $f_\theta(g \cdot x) = \pi_g \cdot f_\theta(x)$ up to the layer normalisation.

What is exact and what is learned. Proposition 1 is a statement about *patch geometry*, and it holds regardless of the network. It is not a claim that our encoder is equivariant. A plain ViT is not: its learned positional embeddings are not D_4 -symmetric, so equivariance is the target that

$\mathcal{L}_{D_4}$ optimises towards rather than a property the architecture enforces, unlike hard-wired group-equivariant networks [Cohen and Welling, 2016]. The measured residual is small but nonzero: the term falls from 1.9×10^{-2} at epoch 1 to 1.5×10^{-4} at epoch 100 (Figure 5).

Proposition 2 (The ring descriptor is D_4 -invariant by construction). *Let $R_k = \{q : \lfloor \rho(q)K \rfloor = k\}$ be the ring partition, where $\rho(q)$ is the distance from q to the grid centre normalised by the half diagonal. Then $\pi_g(R_k) = R_k$ for every $g \in D_4$ and every k .*

Proof. Both generators computed above are isometries of the grid that fix its centre $(\frac{G-1}{2}, \frac{G-1}{2})$: π_r is a rotation by a quarter turn about that point and π_s a reflection in a line through it. Hence $\rho(\pi_g(q)) = \rho(q)$ for all q , so π_g preserves each level set of $\lfloor \rho K \rfloor$. $\square$

Consequently ϕ_k pools over a set that the group action leaves fixed, so matching ϕ across views adds a constraint on *arc morphology at each radius* rather than re-imposing the orientation constraint that $\mathcal{L}_{D_4}$ already carries. The two priors are therefore not redundant.

Numerical validation. All four claims above are checked directly against the training code by `scripts/validate_theory.py`, which verifies, for $G = 10$ and $P = 16$: (5) pixel-for-pixel for all eight group elements; that each π_g is a bijection and π_e the identity; that $\pi_a \circ \pi_b = \pi_{ab}$ for all 64 pairs; and that the ring partition of Proposition 2 is invariant under all eight elements. All pass exactly.

Measured on one RTX A4500, the two lensing terms add 6% to an optimisation step, 202 against 191 ms. Our teacher-free step is dearer than I-JEPA’s, 202 against 80 ms, because it runs two full-image encoder passes with gradients where I-JEPA runs one masked context pass and a no-grad teacher. I-JEPA needs $3 \times$ the epochs, however, so the totals are comparable: 7.7 hours against 9.1.

Table 2: **Measured pretraining cost**, one RTX A4500, batch 64, median over 12 interleaved rounds with the repository’s own modules. Epoch and total times extrapolate the measured step to the 87.5k-image Model I epoch at the epoch budget each method actually used. No wall-clock time was recorded during the original runs, so these were measured separately.

Objective	Epochs	ms / step	min / epoch	Total (h)
LeJEPA	100	191	4.4	7.3
Lens-LeJEPA (ours)	100	202	4.6	7.7
I-JEPA	300	80	1.8	9.1

F Additional classification results

This section reports everything the main text summarises, so that no comparison is hidden by the selection used in Figure 2.

Representation quality without an adapter. Table 3 isolates the encoders from the adaptation method: a frozen linear probe has no capacity of its own, so it measures the representation directly. Only the 300-epoch reference sweep was run this way, and neither matched-budget encoder has a probe run, so this table compares I-JEPA against the Lens-JEPA family rather than against our own encoder. Two things stand out. Every probe is far below what the same encoder reaches with an adapter, I-JEPA going from 0.6002 to 0.9780 at full data, which is why we report adapters throughout. And the ordering under a probe is not the ordering under an adapter: Lens-JEPA-Sym leads the probe table at four of six settings, yet Table 6 puts I-JEPA ahead of it by more than ten accuracy points once adapters are allowed. Probe rankings alone would have been misleading here.

The complete few-shot matrix. Table 6 adds every model at every budget, including the ones Figure 2 omits. Read honestly it says: I-JEPA at $3 \times$ the pretraining budget is ahead of Lens-LeJEPA between 500 and 5,000 labels, and LensPINN, which is supervised and physics-informed, is ahead of everything at every budget. Our claim is specifically about the matched-budget comparison and about the low-label end.

Table 3: **Frozen linear probe**, held-out accuracy, no adapter. Best in each column is bold. Only the $3\times$ -budget reference encoders were run under this protocol.

Encoder	Pretrain epochs	Model II accuracy by label budget					Model III
		100	500	1k	5k	full	full
I-JEPA	300	0.3848	0.4121	0.4351	0.4884	0.6002	0.6235
Lens-JEPA	300	0.3511	0.4101	0.4754	0.5460	0.6312	0.5883
Lens-JEPA-Sym	300	0.3710	0.4489	0.4687	0.5551	0.6443	0.6072

Table 4: **How to adapt a frozen lensing encoder**. Held-out accuracy under seven adapter families and full fine-tuning. rsLoRA is best on all six settings and matches full fine-tuning to within 1.2 accuracy points while training 11% of the weights.

Adaptation ($r=32$ unless noted)	Trainable	Model II accuracy by label budget					Model III	
		100	500	1k	5k	full	full	Mean
rsLoRA [Kalajdzievski, 2023]	2.36M	0.4815	0.7188	0.7972	0.8989	0.9768	0.9744	0.8079
LoRA+ [Hayou et al., 2024]	2.36M	0.4797	0.7077	0.7903	0.8877	0.9589	0.9646	0.7981
LoRA [Hu et al., 2022]	2.36M	0.4763	0.6927	0.7797	0.8809	0.9423	0.9453	0.7862
DoRA [Liu et al., 2024]	2.39M	0.4728	0.7044	0.7720	0.8842	0.9407	0.9409	0.7858
gated LoRA [Zhang et al., 2023]	2.36M	0.4751	0.7003	0.7775	0.8759	0.9426	0.9425	0.7856
layerwise-rank LoRA	1.77M	0.3757	0.6943	0.7756	0.8817	0.9441	0.9529	0.7707
LoHA [Hyeon-Woo et al., 2022]	4.72M	0.3592	0.5807	0.7561	0.8694	0.9192	0.9295	0.7357
full fine-tuning (no adapter)	21.4M	0.4851	0.7150	0.8084	0.8979	0.9751	0.9776	0.8098

Table 5: **LoRA rank**, chosen on the Model II *validation* split only and then fixed for every reported run. These are validation numbers and are not comparable to the held-out test tables.

LoRA rank r (with $\alpha = 2r$)	2	4	8	16	32	64
Trainable parameters	0.15M	0.29M	0.59M	1.18M	2.36M	4.72M
Model II <i>validation</i> accuracy	0.9248	0.9528	0.9663	0.9723	0.9814	0.9604

Table 6: **Complete Model II few-shot matrix**, including every reference encoder and every supervised baseline, so nothing that appears in the main text is selected for favourable comparison. I-JEPA at a $3\times$ pretraining budget leads our encoder between 500 and 5,000 labels, and LensPINN leads everywhere.

Method	Accuracy					Mean one-vs-rest AUROC				
	100	500	1k	5k	full	100	500	1k	5k	full
Lens-LeJEPA + rsLoRA (ours)	0.4815	0.7188	0.7972	0.8989	0.9768	0.6668	0.8475	0.9016	0.9604	0.9927
Lens-LeJEPA + LoRA (ours)	0.4763	0.6927	0.7797	0.8809	0.9423	0.6530	0.8457	0.8877	0.9664	0.9785
Lens-LeJEPA, full fine-tuning (ours)	0.4851	0.7150	0.8084	0.8979	0.9751	0.6721	0.8539	0.9140	0.9616	0.9932
LeJEPA + LoRA	0.3659	0.6251	0.7333	0.8688	0.9256	0.5317	0.7984	0.8702	0.9585	0.9861
I-JEPA + LoRA	0.4311	0.8461	0.8963	0.9311	0.9780	0.6275	0.9478	0.9752	0.9900	0.9981
Lens-JEPA-Sym + LoRA	0.4152	0.5390	0.6494	0.7969	0.8660	0.6034	0.7017	0.8144	0.9232	0.9652
Lens-JEPA + LoRA	0.3468	0.4353	0.4957	0.6792	0.8298	0.5301	0.6148	0.6747	0.8485	0.9428
LensPINN (supervised)	0.9097	0.9843	0.9945	0.9987	0.9997	0.9849	0.9995	0.9999	1.0000	1.0000
ResNet18 (supervised)	0.5999	0.8955	0.9863	0.9979	0.9999	0.7792	0.9787	0.9989	0.9999	1.0000
Vitformer (supervised)	0.6301	0.7540	0.9055	0.9401	0.9820	0.7878	0.8829	0.9706	0.9879	0.9977
ViT (supervised)	0.4989	0.7918	0.8321	0.8829	0.9397	0.6732	0.9207	0.9435	0.9687	0.9934
ViT-SD (supervised)	0.4243	0.7781	0.8305	0.8426	0.8960	0.5831	0.9081	0.9392	0.9483	0.9774

408 **Which class is actually hard.** Table 7 splits the AUROC by class. The no-substructure class is
409 nearly saturated for every method, so the aggregate metric is dominated by the axion-versus-CDM
410 distinction, which is the physically interesting one: both add small-scale structure to the arc and differ
411 in its character rather than its presence. This is where the lensing priors help most.

412 **Adapters and rank in detail.** The next three figures expand the ablation summarised in Section 4.4:
413 the rank sweep and the adaptation regimes, then the full adapter-by-budget grid, then the held-out
414 ROC and confusion structure of the selected configuration.

Table 7: **Per-class one-vs-rest AUROC at full data.** The no-substructure class is easiest for every method. The hard pair is axion versus CDM, and this is where the lensing priors help most: rsLoRA raises the axion AUROC from 0.9694 to 0.9899 on Model II relative to standard LoRA on the same encoder.

Method	Model II one-vs-rest AUROC			Model III one-vs-rest AUROC		
	axion	CDM	no subs.	axion	CDM	no subs.
Lens-LeJEPa + rsLoRA (ours)	0.9899	0.9885	0.9998	0.9910	0.9896	0.9996
Lens-LeJEPa + LoRA (ours)	0.9694	0.9669	0.9992	0.9685	0.9636	0.9992
LeJEPa + LoRA	0.9825	0.9774	0.9984	0.9760	0.9722	0.9983
I-JEPa + LoRA	0.9972	0.9971	0.9999	0.9983	0.9980	0.9998
Lens-JEPa + LoRA	0.9391	0.8975	0.9920	0.9434	0.8768	0.9769
LensPINN (supervised)	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ResNet18 (supervised)	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

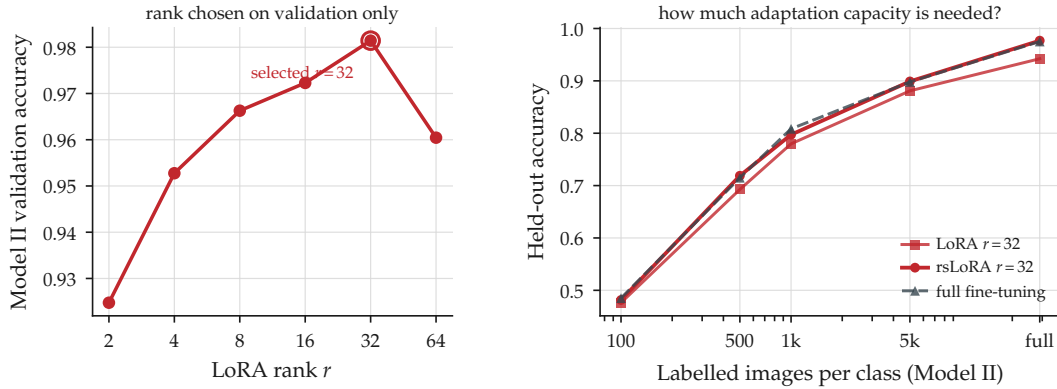


Figure 6: **Adaptation ablations.** Left: the LoRA rank search, run on the Model II validation split only, peaks at $r=32$ and degrades at $r=64$. Right: rsLoRA tracks full fine-tuning across the whole label range while training 11% of the weights, and both are far above a frozen linear probe on the same features.

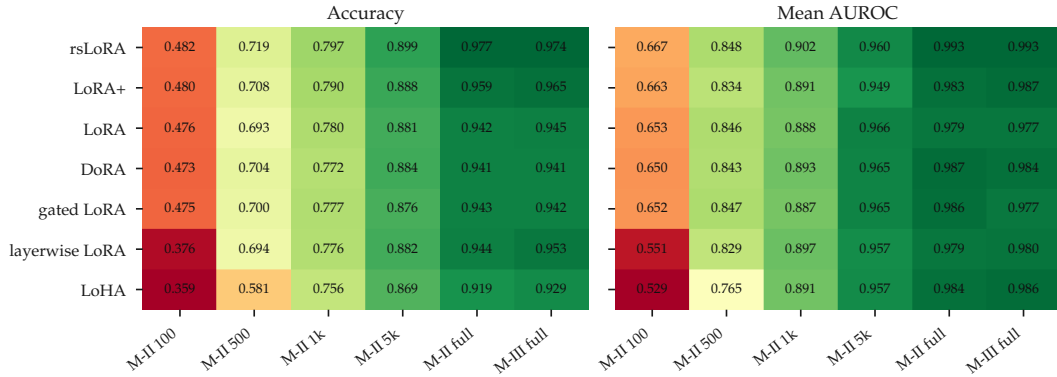


Figure 7: **Adapter family by label budget,** accuracy (left) and mean AUROC (right) for the frozen Lens-LeJEPa encoder. rsLoRA is the most consistent choice. The variants that reparameterise the update more aggressively (LoHA, layerwise) are the ones that break down at 100 labels.

415 G Additional regression and super-resolution results

416 The main text reports one metric per task for space. Tables 8 and 9 give the rest, and the figures that
 417 follow show where the errors actually sit, which is more diagnostic than the aggregates.

418 **Where the regression error lives.** Figure 9 shows that the frozen encoder tracks the target across
 419 the full mass range rather than failing in one region, but with visibly wider scatter than the supervised

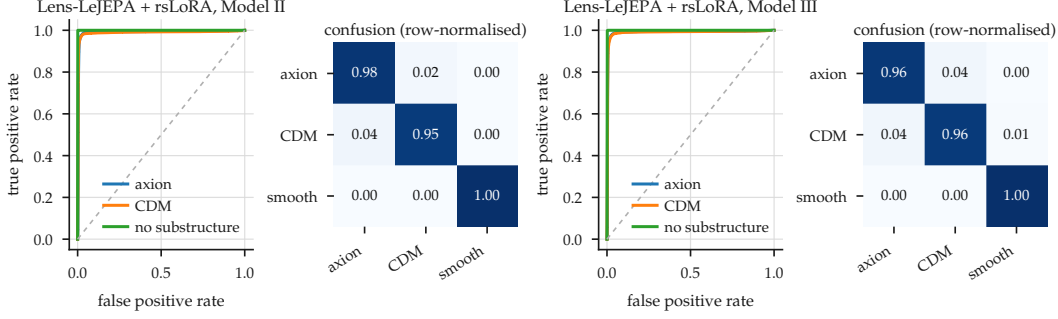


Figure 8: **Held-out ROC curves and confusion matrices** for Lens-LeJEPA + rsLoRA at full data. The residual confusion is almost entirely between axion and CDM; the smooth no-substructure class is separated cleanly on both instruments.

Table 8: **Axion mass regression, all metrics.** The compact version in the main text reports MAE only.

Method	Model II			Model III		
	MAE (dex) ↓	RMSE (dex) ↓	R^2 ↑	MAE (dex) ↓	RMSE (dex) ↓	R^2 ↑
Lens-LeJEPA + rsLoRA (ours)	0.1001	0.1443	0.9374	0.0899	0.1265	0.9524
Lens-LeJEPA + LoRA (ours)	0.1487	0.2024	0.8768	0.1086	0.1505	0.9325
LeJEPA + LoRA	0.1213	0.1628	0.9203	0.1223	0.1614	0.9224
LensPINN (supervised)	0.0423	0.0633	0.9880	0.0351	0.0485	0.9930
ResNet18 (supervised)	0.0209	0.0374	0.9958	0.0192	0.0290	0.9975

Table 9: **Synthetic $2\times$ super-resolution, all metrics.** The compact version in the main text reports PSNR only.

Method	Model II			Model III		
	PSNR (dB) ↑	SSIM ↑	MAE ↓	PSNR (dB) ↑	SSIM ↑	MAE ↓
LeJEPA + LoRA	48.32	0.9996	0.0007	45.38	0.9990	0.0010
Lens-LeJEPA + LoRA (ours)	44.70	0.9976	0.0014	42.50	0.9981	0.0016
Bicubic upsampling (no training)	41.07	0.9987	0.0019	40.27	0.9982	0.0020
RCAN (supervised, same 50-epoch budget)	19.49	0.7830	0.0308	19.01	0.6914	0.0395

420 network, and that the residual grows monotonically towards the highest mass deciles. That is the
421 regime where the vortex interference pattern is finest, and therefore the regime most affected by the
422 blur invariance the objective enforces.

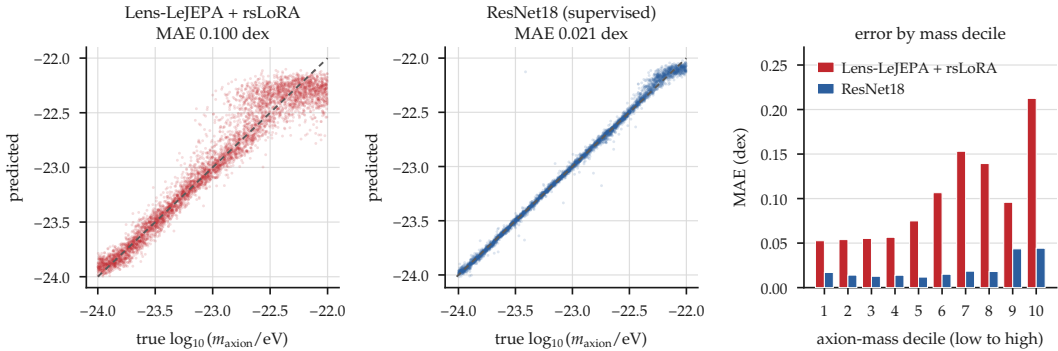


Figure 9: **Axion mass regression diagnostics on Model II.** The frozen encoder tracks the target over the full range but with visibly wider scatter than the supervised network, and the residual grows towards the highest mass deciles where the vortex pattern is finest, the regime most affected by the blur invariance the pretraining objective enforces.

423 **Super-resolution across both instruments.** The last two figures give the metric bars for both test
 424 sets and a second qualitative panel on Model III, whose sharper arcs leave slightly larger residuals
 425 than Model II.

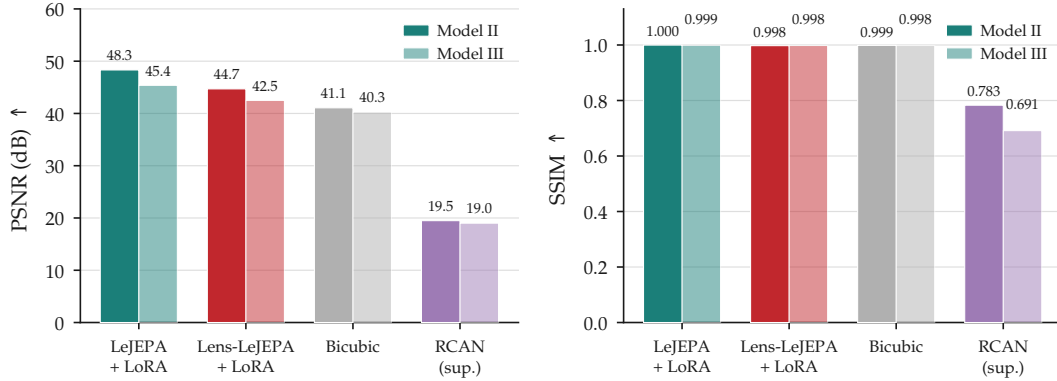


Figure 10: **Super-resolution metrics on both test sets.** Frozen self-supervised features with a light decoder beat bicubic interpolation on PSNR and SSIM for both instruments, while an RCAN trained from scratch under the same 50-epoch budget does not converge.

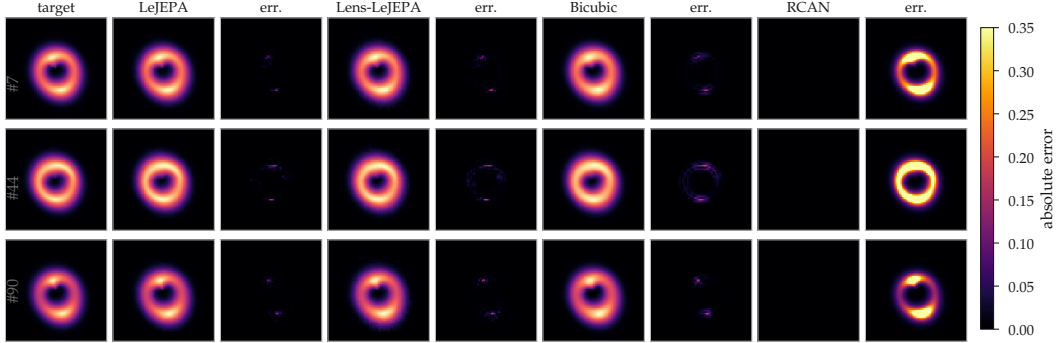


Figure 11: **Super-resolution examples on held-out Model III lenses**, in the same format as the Model II panel of Figure 3. The behaviour matches Model II, with slightly larger residuals reflecting the sharper HST-like arcs.

426 H Limitations

427 All three datasets are `lenstronomy` realisations with a known point spread function, known noise
 428 and a balanced, complete label set. Real survey images differ in all three respects, and additionally
 429 carry selection effects, blending, and lens galaxies whose light must be subtracted. Nothing here
 430 tests that transfer. This bounds every claim in the paper, and it is the obvious next step: the natural
 431 experiment is to pretrain on unlabelled survey cutouts and evaluate on the small set of confirmed real
 432 lenses. Proposition 1 makes the patch permutation exact, but the ViT’s positional embeddings are
 433 not D_4 -symmetric, so the encoder is only approximately equivariant at convergence, with a small
 434 residual (Figure 5). A hard-equivariant backbone would remove that gap and is untested here.

435 I Reproducibility

436 Every number in this paper is read directly from the recorded held-out evaluation files by using the
 437 code provided.